

Tian Bai

Ph.D. | UESTC

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► SUMMARY

I received my Ph.D. degree in computer science from the University of Electronic Science and Technology of China (UESTC) in June 2024 under the supervision of [Prof. Mingyu Xiao](#). My Ph.D. dissertation is entitled "[Investigations Concerning Parameterized and Exact Algorithms for the Feedback Set and Related Problems](#)". I am an EATCS member and CCF member, and my research interests include graph algorithms, approximation and parameterized algorithms, and algorithmic game theory.

► EMPLOYMENT

University of Bergen, Department of Informatics,
Postdoctoral Research Fellow 2025.09 - Present

- Host: Prof. Fedor Fomin.
- Research Interests: Parameterized Algorithms, Prophet Inequality.

The University of Hong Kong, Department of Computer Science,
Postdoctoral Research Fellow 2024.09 - 2025.08

- Host: Prof. Zhiyi Huang.
- Research Interests: Parameterized Algorithms, Contract Design, Prophet Inequality.

► EDUCATION

University of Electronic Science and Technology of China, School of Computer Science and Engineering,
Computer Science, *Ph.D.* 2024.06

- Advisor: Prof. Mingyu Xiao.
- Research Interests: Graph Algorithms, Parameterized Algorithms, Mechanism Design.

University of Electronic Science and Technology of China, BS Class (Math & Physics),
Fundamental Sciences (Mathematics and Physics), *B.Sc* 2016.06

- Research Interests: Algebraic Coding Theory, Cryptography.

► RESEARCH PUBLICATIONS

Journal Articles

- [1] **Tian Bai**, Mingyu Xiao: [A Parameterized Algorithm for Subset Feedback Vertex Set in Tournaments](#). Theoretical Computer Science 975: 114139 (2023).
- [2] **Tian Bai**, Mingyu Xiao: [Exact Algorithms for Restricted Subset Feedback Vertex Set in Chordal and Split Graphs](#). Theoretical Computer Science 984: 114326 (2024).
- [3] **Tian Bai**, Mingyu Xiao: [Computational Complexity of Feedback Vertex Set and Subset Feedback Set Problems: A Survey](#). Journal of computer research and development (2024).
- [4] **Tian Bai**, Mingyu Xiao: Exact Algorithms for Maximum Independent Set Problem on Hypergraphs. SCIENTIA SINICA Informationis. The first non-trivial exact algorithm for a price-collecting version of MIS on Hypergraphs, which breaks the 2^n -barrier.
- [5] **Tian Bai**, Mingyu Xiao: [Solving Subset Feedback Vertex Set in Chordal Graphs Faster than \$2^k\$](#) . Information and Computation (Under Submission)
- [6] Mengfan Ma, **Tian Bai**, Mingyu Xiao, Bakh Khousainov: [Facility Location Games Beyond Single-Peakedness: the Entrance Fee Model](#). Mathematics of Operations Research (Under Submission)

Conference Proceedings

- [1] Junqiang Peng, **Tian Bai**, Jingyang Zhao, Mingyu Xiao: Sustained Vertex Cover on Temporal Graphs. Proceedings of the 35th ACM on Web Conference (WWW) 2026.
- [2] Yingying Huangfu, **Tian Bai**: [A Cooperative Statistical Approach for Abnormal Node Detection with Adversary Resistance](#). Proceedings of the 34th International Conference on Computer Communications and Networks (ICCCN) 2025: 1-6.
- [3] **Tian Bai**, Mingyu Xiao: [Breaking the Barrier \$2^k\$ for Subset Feedback Vertex Set in Chordal Graphs](#). Proceedings of the 49th International Symposium on Mathematical Foundations of Computer Science (MFCS) 2024: 15:1-15:18.

- [4] Mengfan Ma, **Tian Bai**, Mingyu Xiao, Xin Chen: [Facility Assignment with Fair Cost Sharing: Equilibrium and Mechanism Design](#). Proceedings of the 30th International Computing and Combinatorics Conference (COCOON) 2024: 203-215. I also designed unanimous and strategyproof mechanisms for this model.
- [5] Mengfan Ma, Mingyu Xiao, **Tian Bai**, Bakh Khousainov: [Facility Location with Entrance Fees](#). Association for the Advancement of Artificial Intelligence (AAAI) 37(5), 5797-5804. 2023: 5797-5804.
- [6] **Tian Bai**, Mingyu Xiao: [Exact and Parameterized Algorithms for Restricted Subset Feedback Vertex Set in Chordal Graphs](#). Proceedings of the 17th Annual Conference on Theory and Applications of Models of Computation (TAMC) 2022: 249-261.

► TALKS

Presentation: CCF National Conference of TCS (NCTCS) 2025	2025.07
Kernel for Subset Feedback Arc Set in Tournaments.	
Presentation: CCF National Conference of TCS (NCTCS) 2023	2023.07
Computational Complexity of Feedback Vertex Set and Subset Feedback Set Problems: A Survey.	
Presentation: TAMC 2022	2022.09
Exact and Parameterized Algorithms for Restricted Subset Feedback Vertex Set in Chordal Graphs.	

► SERVICES AND PROFESSIONAL ACTIVITIES

Reviewer for Conferences

- ISAAC 2024
- MFCS 2024
- IPEC 2025
- COCOON 2025
- SOFSEM 2025
- WALCOM 2022 2024 2025 2026.

Reviewer for Journals

- Theoretical Computer Science.
- Journal of Computer and System Sciences.
- Discrete Applied Mathematics.
- Frontiers of Computer Science.

Moderator for Workshops

- The 19th International Conference and Workshops on Algorithms and Computation (WALCOM), 2025.
- CCF National Conference of Computational Economics, 2024.
- Algorithms & Logic Workshop at UESTC, 2023.
- Algorithmic Game Theory and Computation Theory Forum at UESTC, 2022.
- The 2nd Theoretical Computer Science Outstanding PhD Students Forum (TCSPhd2021), 2021.

► TEACHING

Teaching Assistant for graduate students: Frontier Topics in Doctoral Research	2022-2023
Teaching Assistant for graduate students: Advanced Algorithm Analysis and Design	2022-2023
Teaching Assistant for undergraduate students: Frontiers of Theory of Computation	2022-2023
Teaching Assistant for undergraduate students: Frontiers of Theory of Computation	2021-2022
Teaching Assistant for undergraduate students: Advanced Algebra	2017-2018
Teaching Assistant for undergraduate students: Linear Algebra	2016-2017

► AWARDS AND HONORS

Outstanding Teaching Assistant	2024
Global 69th/China 10th place in IEEE Global Extreme Programming Competition 12.0	2018
First prize in the National University Mathematics Competition	2014 & 2015